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A Large Language Model-Based Agent for Automated Bidding Strategy Generation in Electricity Markets

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***Abstract*—As electricity markets become increasingly complex due to evolving market structures and the integration of renewable energy, optimizing bidding strategies for power generators has become essential to ensure competitiveness and efficiency. However, traditional bidding methods often struggle to adapt to market fluctuations and the inherent uncertainty of renewable energy generation. To address these challenges, we explore the potential of Large Language Models (LLMs) in developing intelligent and adaptive bidding strategies. This paper proposes a new agent for automated bidding strategy generation with systematic prompt engineering adopting the Chain of Thought (CoT) and auto-debug mechanism based on reflection. Experimental results demonstrate that our approach improves the accuracy, adaptability, and efficiency of automated bidding strategy generation in dynamic electricity markets. This paper highlights the potential of LLMs in energy trading, paving the way for more intelligent and autonomous decision-making agents.**

***Keywords*—large language model, bidding strategy generation, electricity market, prompt engineering, agent**

I. Introduction

Electricity markets are central to the modern energy system, and their efficient and stable operation is crucial for socioeconomic development. Electricity plays a pivotal role in advancing modern economies due to its substantial market and non-market advantages [1]. In power system operation, the primary goal is to minimize the total economic costs of maintaining system stability while ensuring supply-demand equilibrium [2]. As the core mechanism of electricity market operations, generator bidding strategies impact resource allocation and market efficiency. Efficient bidding strategies not only simplify the market-clearing process but also enhance the profits of power producers [3]. As a key component of the energy transition, electricity markets are undergoing a fundamental transformation driven by decarbonization targets and the increasing penetration of renewable energy sources [4]. The increasing penetration of renewable energy introduces both new opportunities and challenges, particularly in electricity market operation.

Modern electricity markets are becoming more complex, with significant uncertainty and non-linearity. Traditional bidding strategies, which rely on fixed rules and parameters, struggle with changing conditions. Moreover, the intermittent and unpredictable nature of renewable resources such as wind and solar power increases uncertainty in power system operation [5]. This makes generator bidding decisions more difficult, as traditional models cannot handle such volatility. These limitations reduce strategic competitiveness and lower market efficiency.

Strategic bidding in electricity markets is an optimization problem traditionally solved using heuristic and model-based methods. However, these methods face challenges due to the non-differentiability of objective functions and constraints, as well as their sensitivity to parameter selection [6]. Traditional methods, such as rule-based and cost-based strategies, are often iterative, making them inefficient and impractical for real-world applications [7]. While heuristic methods, game-theoretic approaches, and reinforcement learning have been explored for optimal bidding strategies, they struggle to accurately model the complex dynamics of markets and appropriately handle real-world uncertainties [8]. The challenges, such as reliance on manual adjustments, difficulty in modeling dynamic market

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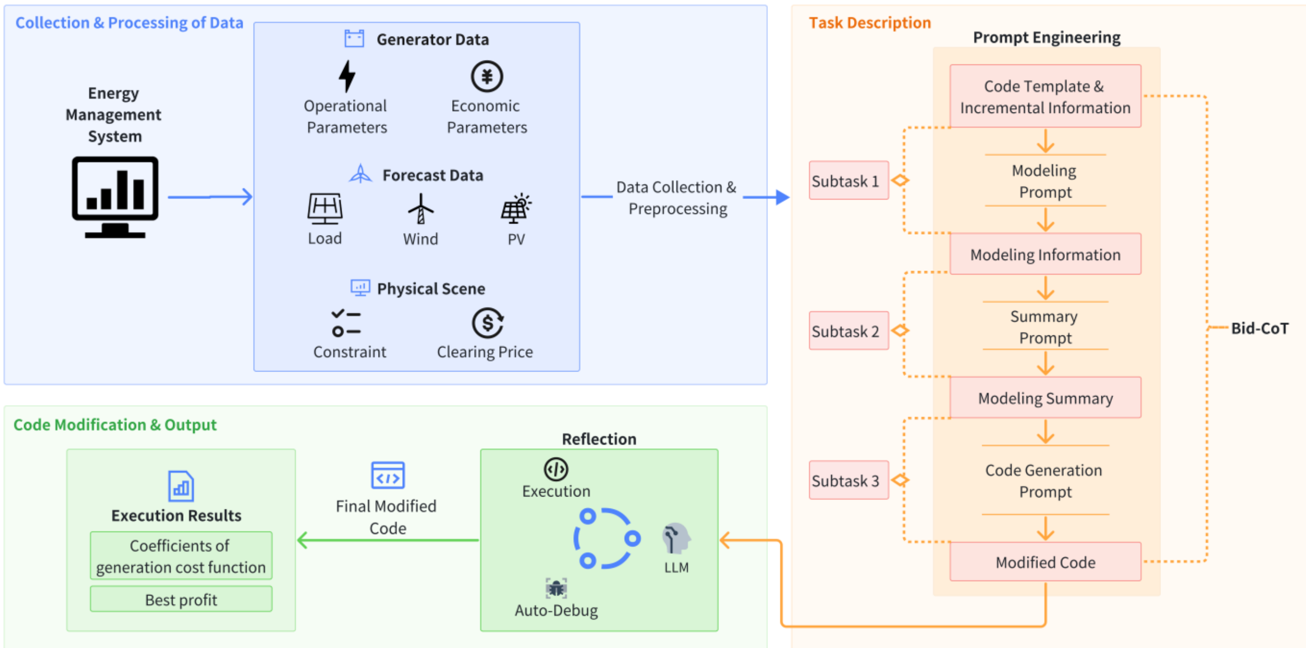

Fig. 1. LLM-based agent of automated bidding strategy generation.

behavior, and parameter optimization issues, reduce their practical utility in dynamic markets. These methods often lack the flexibility and intelligence needed to handle the increasing complexity and uncertainty of modern electricity markets.

Few methods comprehensively address the challenges of renewable energy, including its distributed nature, intermittency, and multi-objective real-time coordination. To fill these gaps, large language models (LLMs), an advanced AI technology, offer significant potential by enabling intelligent agents to cooperate, compete, and make strategic decision, with their strong capabilities in knowledge acquisition, reasoning, and human interaction [9]. Through analyzing historical energy demand data, weather forecasts, and news, LLM assists power companies in predicting future demand, allowing for proactive optimization of supply strategies [10]. Therefore, a fully automated LLM-based agent is assumed to optimize bidding strategies dynamically, improving efficiency and adaptability in electricity markets.

In power systems, LLMs, including GPT models, have proven valuable for tasks such as fault diagnosis, predictive maintenance, and real-time grid optimization [11]. To optimize bidding strategies effectively, LLMs can be integrated into autonomous agents, creating LLM-based systems that interact dynamically with market environments, make strategic decisions, and adapt to real-time changes. Additionally, by employing prompt engineering with carefully designed prompts, LLMs enhance their performance and applicability across various domains [12]. Chain of Thought (CoT) method, as a prompting technique, decomposes complex problems into interpretable step-by-step solutions, improving interpretability and reasoning [13]. Incorporating prompt engineering and structured problem decomposition into the bidding agent, it is possible to develop a systematic prompt engineering methodology specialized in strategy optimization for market bidding, which may better capture the complexity and dynamics of electricity markets. Besides, the reflection mechanism enables autonomous error correction through iterative self-evaluation [14]. With iterative self-evaluation and debugging, the agent refines the bidding strategy and further optimizes generated code, ensuring robustness and reliability.

Based on the application of LLMs in electricity markets' bidding strategies, the contributions of this paper are as follows:

1. We propose a fully automated agent based on LLMs to address challenges caused by renewable energy fluctuations and regional complexity in electricity markets, significantly improving decision-making efficiency.
2. We introduce Bid-CoT, a structured prompt glossary combined with CoT for bidding strategies. This approach improves the accuracy and reliability of the code generated by LLMs.
3. We design the auto-debug mechanism for code based on reflection, enabling continuous code refinement through autonomous error detection, reflection summary and automatic debugging by LLMs.

II. Automated Bidding Strategy Generation Based on Large Language Model

A. Overall Agent

We introduce an agent with a specialized prompt engineering adopting Bid-CoT, empowered by LLMs to improve the analysis and optimization for electricity markets' bidding strategies. The agent consists of three modules: collection and processing of data, task description, and code modification and output. The agent is shown in Fig. 1.

Firstly, the Energy Management System (EMS) inputs data related to generator bidding, including generator data, forecast data and physical scene information. This data is collected and

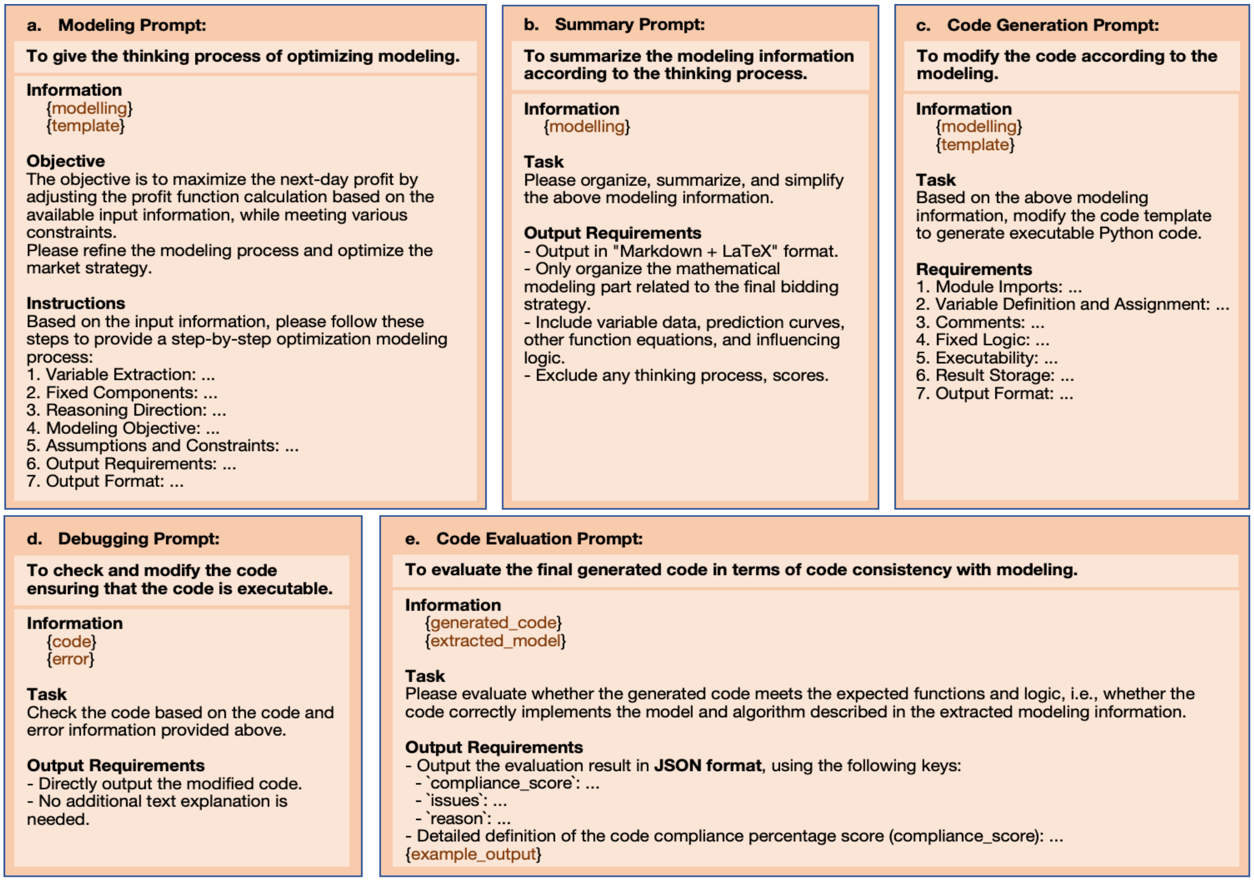

Fig. 2. Systematic prompt engineering.

preprocessed for subsequent tasks. Generator data encompasses operational parameters including maximum power and minimum power, and economic parameters including various kinds of generation costs. In our day-ahead electricity market bidding scenario, the agent will simulate a coal-fired unit body with the goal of maximizing profit by adjusting the power generation cost function coefficients. Accurate cost function calculations are crucial for dynamic market decision-making, particularly in determining day-ahead electricity market profits. In particular, fuel cost and carbon emission cost require both their respective average intensity curves and fuel prices and carbon prices as sources of input data. Additionally, the simulated coal-fired generators need to account for the forecast information on renewable energy, to adjust the market price in a timely and flexible manner. Forecast data includes 24-hour hourly variations in load, wind and photovoltaic (PV) forecasts to simulate the output fluctuations of renewable energy. In the specific electricity market decision-making scenario, we also should meet the constraint conditions of physical scenes, as well as the historical clearing price range and reserve market clearing price.

Subsequently, we incorporate the information of the processed data into the prompts of task to purposefully generate the modified code. The agent utilizes structured prompt engineering to systematically optimize bidding strategies and modify code templates. In this agent, our Bid-CoT decomposes the task into a series of sub-tasks. Guided by interrelated prompts designed for sub-tasks, the LLM-based agent ends up generating the adjusted code based on the instructions for the target, ensuring logical consistency and adaptive decision-making in response to market fluctuations.

The last module performs auto-debug on the modified code outputted by LLM, to obtain the improved code and its execution results. This operation can further enhance the reliability and effectiveness of the ultimate code. After going through the mechanism of LLM-driven reflection, the agent outputs the final results based on the debugged code. The results of the final execution, including the adjustable cost function coefficients and the best profit, are stored in the document.

B. *Systematic Prompt Engineering for Bidding Strategy*

Recent advances in LLMs have strengthened their ability to process complex instructions, while prompt engineering further improves their adaptability, enabling better decision-making [15]. To improve the optimization quality for bidding strategy, we introduce Bid-CoT, a structured prompting methodology designed to systematically guide LLMs through the bidding process. Traditional CoT reasoning often lacks structural coherence and adaptability in dynamic scenarios, such as renewable energy fluctuations and complex market conditions. Furthermore, the multi-step reasoning process of standard CoT can be excessively long in certain bidding scenarios, making it challenging to meet the high-frequency decision-making demands of electricity markets while maintaining logical consistency.

Bid-CoT addresses these challenges by decomposing the task into three structured prompts: modeling, summarization, and code generation. This ensures a clear problem formulation, refines key insights, and translates them into executable codes. Instead of sequentially optimizing each step, this approach ensures that intermediate outputs serve as inputs for subsequent prompts, enabling progressive refinement. By structuring the reasoning process in this way, Bid-CoT optimizes strategy formulation while mitigating risks associated with excessively long reasoning chains, such as error accumulation and contextual drift.

- Modeling Prompt: This initial prompt provides the LLM with essential information, such as the code template, incremental input data, decision objectives, and output formats. The agent is instructed to explicitly convey its reasoning process for modeling optimization. This step establishes a systematic and adjustable approach to problem understanding, ensuring a coherent foundation for subsequent tasks.
- Summary Prompt: The second prompt processes the reasoning output from the modeling phase to extract key insights and generate a structured modeling summary. This step guarantees that instructions for code modification remain clear and logically organized, facilitating an efficient and well-informed transition to code generation.
- Code Generation Prompt: Based on the modeling summary and additional critical information, this prompt directs the LLM to modify the initial code template, producing an optimized implementation tailored to the current market scenario.

Beyond the three core prompts, our systematic prompt engineering incorporates two additional prompts to further refine and validate the generated code:

- Debugging Prompt: This prompt provides the LLM with the modified code and execution feedback, enabling iterative refinement to improve correctness and reliability.
- Code Evaluation Prompt: The final prompt evaluates the output code against the target modeling, instructing the LLM to evaluate and score the code based on functionality and logical consistency.

By integrating these systematically structured prompts, Bid-CoT ensures a robust, adaptive, and high-precision approach to bidding strategy optimization. The systematic prompt engineering is shown in Fig. 2.

C. *Reflection Agent*

To improve the quality and reliability of LLM-generated code, we implement a reflection mechanism. Reflection allows the LLM agent to systematically analyze past outputs based on relevant observations, enabling evaluation and iterative improvement [16].

Without a structured reflection mechanism, the quality of the modified code may fall short of expectations, and the executability of the output code may not be reliably ensured. To

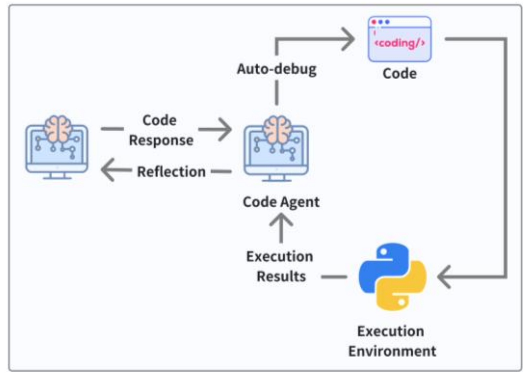

Fig. 3. Reflection agent.

efficiently address errors arising during code execution, we integrate reflection-driven auto-debugging, which refines the code within a constrained number of iterations. This approach not only enhances debugging efficiency but also significantly improves the quality and reliability of the final code. The detailed structure of the reflection agent is illustrated in Fig. 3.

III. EXPERIMENT

A. *Data*

This paper uses publicly available IEEE 300-bus system test cases to get adequate coal-fired generator data. The missing values and unknown parameters concerning generator sets are filled with default values based on empirical data, with random noise added due to uncertain environmental factors.

Along with operational and economic parameters of generation units, the input data includes 24-hour forecast data for load, wind and PV output, and other relevant physical scenario information such as historical electricity market clearing price ranges and reserve market clearing prices. Load, wind and PV forecasts are derived from the time-series day-ahead data used to run PCM simulations in the Reliability Test System Grid Modernization Lab Consortium (RTS-GMLC), providing hourly resolution time-series information. Load data uses hourly forecast electricity demand for several specific areas. Wind data and PV data contain the hourly forecast generation from wind plants and utility-scale PV plants. The raw data is cleaned to ensure data quality and consistency, subsequently transformed to a suitable format for analysis, like creating a time index to align the data from different sources and facilitate time-based calculations.

After processing a series of day-ahead data, we get multiple sets of 24-hour forecast information for determining forecast curves. In each group of comparative experiments, we will set up 3 parallel tests with different sets of forecast information input, under the premise of ensuring the consistency of other factors. Each individual test has ten sets of sequentially extracted 24-hour forecast data to calculate the final measurements. And we will finally take the average of results from parallel tests concerning every evaluation indicator as the record for experimental comparison.

B. *Test Criteria*

To measure the quality of LLM-generated code, we analyze the experimental results from three metrics: success rate, effectiveness rate, and code & model consistency score. The specific details are as follows:

1) Success Rate: The success rate of code execution, as the most basic metrics to evaluate the quality of the final modified code, measures the ratio of the number of successful final executions to the total number of input sets, mainly reflecting whether the code is correct and can be executed. If the success rate is low, it indicates that the code generated by the current method has many syntax or logic errors, requiring adjustments to the LLM prompt and debugging approach. The calculation formula is as follows:

$$R_S = \frac{\#\text{Successful Final Executions}}{\#\text{Total Input Sets}} \times 100\% \qquad (1)$$

2) Effectiveness Rate: The effectiveness rate measures the extent to which the generated code solves the given problem or produces new, useful results. It evaluates the practical utility of the generated code. If the ratio is low, the code generated by the LLM does not reach the desired effect and does not achieve the task purpose of improving the strategy. We calculate this proportion through the following formula:

$$R_E = \frac{\#\text{Results Different from the Initial}}{\#\text{Total Input Sets}} \times 100\% \qquad (2)$$

3) Code & Model Consistency Score: We use prompt words that specify the code evaluation and score output in JSON format to guide the LLM to score the compliance of the output code with the target modeling, and then take the average of a set of scores. This metric reflects whether the generated code meets the modeling goals and constraints identified during the thinking process, measuring the accuracy of the LLM's understanding of the model. If this consistency rate is low, the generated code does not adequately match the assumptions of the model. The final average is calculated by the following formula:

$$S_C = \frac{1}{n}\sum_{i=1}^{n} \text{Compliance Score}_i \qquad (3)$$

C. LLM Agent Performance in Bidding Strategy Generation

To evaluate the performance of our LLM-based agent in optimizing bidding strategies within dynamic electricity markets, we conduct comparative experiments using Gemini and DeepSeek model.

TABLE I. PERFORMANCE COMPARISON OF LLMS

	Gemini	DeepSeek
Success Rate	**100.00%**	100.00%
Effectiveness Rate	**100.00%**	100.00%
Code & Model Consistency Score	**90.17**	79.17

The experimental results in Table I compare the performance of the Gemini and DeepSeek models. Gemini's success rate and its effectiveness rate are both 100.00%, and its Code & Model Consistency Score is 90.17. In contrast, DeepSeek also attains a success rate and effectiveness rate of 100%, but its Code & Model Consistency Score is 79.17, which is, on average, lower than that of the Gemini model. These results suggest that both models perform well overall, with Gemini demonstrating superior performance in this scenario. We can deduct from the findings that the more powerful the model, the better the effect, highlighting the generality and potential for further development of the adopted agent.

To further validate the effectiveness of our LLM-based agent in optimizing bidding strategies, we analyze the optimized cost function parameters a *and* b as well as their impact on profit. The bidding cost C follows a quadratic function:

$$C = a \cdot P^2 + b \cdot P \qquad (4)$$

where P represents power output, a is the coefficient of the quadratic term, and b is the coefficient of the linear term in the bidding cost function, both optimized to maximize profit. The constant term is set to zero for simplification in the bidding model.

The overall profit in electricity markets is calculated as:

$$\text{Profit} = \text{Market Revenue} - \text{Bidding Cost} - \text{Actual Cost} \qquad (5)$$

Table II shows the optimized parameters and profit changes obtained using Gemini. The results show a significant increase in a (from 9.99e-07 to 8.00e-04 in most cases), indicating a shift toward a more flexible pricing strategy, while b remains stable around 300. Profit improvements are observed in most cases, with a maximum increase of 12,892,006.86 CNY, demonstrating the effectiveness of the LLM-generated bidding strategy.

TABLE II. EXECUTION RESULTS BEFORE AND AFTER OPTIMIZATION

	Before		After		Profit Variation (CNY)
	a	b	a	b	
1	9.99e-07	299.90	8.00e-04	300.00	12892006.86
2	1.00e-06	299.93	9.99e-07	299.91	1193478.31
3	9.99e-07	299.91	8.00e-04	300.00	589106.54
4	9.99e-07	299.91	8.00e-04	300.00	0
5	9.99e-07	299.88	8.00e-04	300.00	2349271.52
6	1.00e-06	299.88	8.00e-04	300.00	-1374389.18
7	1.00e-06	299.93	9.99e-07	299.91	660300.74
8	1.00e-06	299.93	8.00e-04	300.00	2546508.09
9	1.00e-06	299.95	9.99e-07	299.91	4574388.73
10	9.99e-07	299.88	8.00e-04	300.00	-2997137.42

D. Comparison of Prompt Engineering Implementation

To test the effect of Bid-CoT method, we compare its performance with the approach where the LLM directly generates modified code from a simple prompt without Bid-CoT.

Table III shows the performances of the group with a single prompt and the group under our agent. The group with a single prompt and the group using Bid-CoT both have a success rate of 100.00%. However, the group with a single prompt demonstrates its effectiveness rate to be 63.34%, which is quite lower than the Bid-CoT group. Regarding Code & Model Consistency Score, using single prompt leads to an average value of 49.90, much lower than the result of 90.17 brought by

Bid-CoT. It is clear that the three metrics of the final code obtained by the prompt engineering using Bid-CoT method under our agent are better than the comparison of a single prompt, which shows the rationality of our prompt engineering.

TABLE III. PERFORMANCE COMPARISON OF PROMPT METHODS

	Bid-CoT	Single Prompt
Success Rate	**100.00%**	100.00%
Effectiveness Rate	**100.00%**	63.34%
Code & Model Consistency Score	**90.17**	49.90

E. Comparison of Reflection

To verify the continuous optimization effect of reflection, we design a group without reflection to compare with the original experimental group applying the Gemini model.

TABLE IV. PERFORMANCE COMPARISON OF REFLECTION

	With Reflection	Without Reflection
Success Rate	**100.00%**	96.67%
Effectiveness Rate	**100.00%**	100.00%
Code & Model Consistency Score	**90.17**	89.00

Table IV shows a performance comparison between experiments with and without reflection. Notably, the group with reflection achieves a higher success rate of 100.00% and a greater Code & Model Consistency Score of 90.17, compared to 96.67% and 89.00 for the group without reflection. Both groups show an effectiveness rate of 100%. These results show that auto-debug in reflection is beneficial to improve the quality of generated code and plays an important role in the overall effect of the agent.

IV. CONCLUSION

This paper explores the application of LLMs for optimal modeling, modification, debugging, and evaluation in the context of automated bidding strategy generation for electricity markets. Considering the challenges of renewable integration, diverse generation dynamics, and evolving market mechanisms, our agent incorporates Bid-CoT, a fully automated approach built upon CoT reasoning. It systematically structures prompt engineering and incorporates LLM agents for self-evaluation and refinement. By embedding domain-specific knowledge through multidimensional professional prompts, the agent achieves both revenue maximization for power producers and enhanced decision-making efficiency. It provides a flexible and efficient solution to adapt to the rapid changes in electricity markets, enabling better responsiveness to fluctuations driven by the energy transition and the increasingly complex grid supply and demand. Future research will focus on fine-tuning LLMs and building specialized datasets for specific transactional scenarios. This will lead to more accurate generation of optimal bidding strategies while maintaining the universality of the agent across diverse market contexts.

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